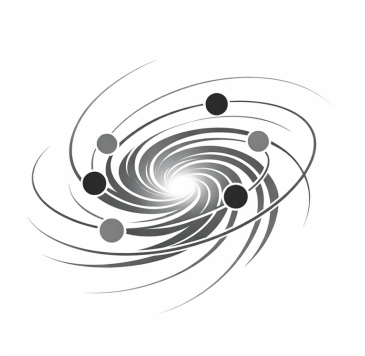


Part of the Monistic Continuum Model (MCM)



Dark Matter (DM)

An MCM Interpretation of Dark Matter

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Abstract

Modern physics describes dark matter as an invisible, non-electromagnetically radiating component that can be observed only through its gravitational effects. Models such as Cold Dark Matter, self-interacting dark sectors, dark radiation, and neutrino-like relics adopt this idea without formulating a continuum-based ontology. The Monistic Continuum Model (MCM) extends this approach and interprets dark matter as residual tensions within a universal medium: short-lived vortex remnants that do not emit radiation and can be detected solely as tension or potential sinks. Thus, the MCM provides a consistent, particle-free explanation for dark matter that is fully compatible with all current observations and expands physical models with an ontologically closed continuum perspective.

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1 Introduction

For several decades, modern physics has attempted to investigate the nature of dark matter both experimentally and theoretically. Numerous research programs—from precision measurements of cosmological structures to gravitational-lensing analyses and direct as well as indirect detection efforts—aim to uncover that invisible component of the universe which accounts for roughly one quarter of the cosmic energy density. Despite these intensive efforts, the material character of dark matter has remained undetermined: no new particles have been detected, nor has any electromagnetic radiation or interaction been observed. Physics therefore describes dark matter as an invisible, non-radiating component that can be measured exclusively through gravitational effects.

At precisely this point, a remarkable proximity to the theory of the Monistic Continuum Model (MCM) becomes apparent. Many physical observations and model assumptions—such as the absence of electromagnetic radiation, purely gravitational visibility, large-scale structural influence, and stability over cosmic timescales—align with the MCM interpretation, in which dark matter is understood as residual vortices of the universal medium. These short-lived, neutral tension remnants arise from the dynamics of matter vortices, emit no radiation, and act solely through tension sinks that manifest on cosmic scales as gravitational potentials.

Thus, the classification of dark matter within the MCM not only provides an ontologically closed alternative to particle-based models but also represents a natural continuation of the properties already observed by physics. Existing physical efforts and measurement results therefore do not contradict the MCM theory; instead, they form an empirical foundation that strongly supports the continuum interpretation of dark matter.

2 Cosmic Ground State in the MCM

The cosmic ground state forms the starting point of all subsequent vortex processes in the Monistic Continuum Model (MCM) and thus the entire material development of the universe. It describes a homogeneous, tension-bearing continuum without material structures, without vortex cores, and without any form of radiation. This state is not static but dynamically under-tensioned: minimal fluctuations in the tension field lead to the formation of the first proto-vortices, initiating the transition to cosmic dynamics.

2.1 Definition of the Continuum

The continuum of the MCM is a unified, non-segmented medium that contains no particles, no fields, and no discrete units. It possesses an intrinsic tension energy, understood as a fundamental property of the medium. This tension is not localized but distributed uniformly throughout the continuum and forms the basis for all later vortex processes.

The continuum is:

- **homogeneous**: no preferred directions or locations,
- **continuous**: no discrete building blocks or particles,
- **tension-bearing**: contains an elastic base tension,
- **dynamic**: minimal fluctuations are always present.

These properties define the cosmic ground state as a medium that is structurally featureless yet still capable of forming structures.

2.2 Tension Field and Fundamental Symmetries

The tension field of the continuum is the central physical quantity of the MCM. It describes the local and global distribution of elastic tension within the medium. In the ground state, this tension field is symmetric and isotropic, meaning it has no preferred direction and no local concentrations.

The fundamental symmetries are:

- **Isotropy**: equal tension in all directions,
- **Homogeneity**: equal tension at all locations,
- **Rotational symmetry**: no preferred axes,
- **Translational symmetry**: no preferred positions.

These symmetries prevent the spontaneous formation of stable structures. Only when fluctuations occur is the symmetry locally broken, making the formation of the first proto-vortices possible.

2.3 Fluctuations and Proto-Vortices

Although the cosmic ground state appears homogeneous, the continuum contains microscopic tension fluctuations. These fluctuations are unavoidable because the medium is dynamic and cannot maintain a perfectly uniform distribution of tension.

If a fluctuation becomes locally strong enough, a **proto-vortex** forms. A proto-vortex is a short-lived, open vortex structure without a core and without a stable rotational geometry. It represents the first step in the development of material vortices but is not yet material in the proper sense.

Proto-vortices are:

- **unstable,**
- **open,**
- **non-quantized,**
- **non-radiating.**

They dissolve quickly and return their tension to the medium. A small fraction of these proto-vortices, however, intensifies and forms the first stable material vortices.

2.4 Why the Ground State Contains No Matter

Matter arises in the MCM exclusively through stable vortex structures with closed axes and quantized rotational tensions. However, the cosmic ground state contains:

- no closed vortices,
- no quantized tension modes,
- no stable rotational axes,
- no flow geometries that could define matter.

Because the tension is distributed homogeneously and no local concentrations exist, no stable vortex structure can form. Only fluctuations break the symmetry and enable the formation of proto-vortices, from which stable material vortices later emerge.

The ground state is therefore a **structureless, tension-bearing initial state** that forms the basis for all later cosmic processes but contains no matter itself.

3 Emergence of Material Vortices

The emergence of material vortices represents the decisive transition from the structureless cosmic ground state to the first stable material configurations in the Monistic Continuum Model (MCM). Local tension fluctuations initially give rise to open vortex structures, which develop through

several intermediate forms into stable material vortices, as shown in the tables. In parallel, residual vortices form as open, non-material tension anomalies that constitute the basis of dark matter in the MCM. This entire development follows clear stability criteria that govern the geometry and dynamics of the continuum.

Property	Open Vortex	Single Vortex	Quark Vortex	Proton Vortex
Ontological	✓	–	–	–
Core	0	1	2	6
Axis	–	✓ (unstable)	✓ (open)	✓ (locked)
Rotation	Tension gradient	Core axis	Core axis	Closed multi-axis
Radiation	–	–	–	✓
Matter	–	–	–	✓
Spectral Signature	–	–	–	✓
Detectable	–	–	–	✓
Stability	Unstable	Unstable	Unstable	Fully stable
Geometry	Open	Open	Open	Closed
Coupling Ability	✓	✓	✓	– (end structure)

3.1 Vortex Formation in the Continuum

The formation of a vortex begins with a local tension fluctuation in the continuum. If this fluctuation becomes sufficiently strong, it breaks the local symmetry of the tension field and a rotating flow geometry emerges. This rotation leads to the formation of an axis that defines the vortex.

A vortex forms when:

- a local tension fluctuation generates a rotational axis,
- the rotational tension is stable enough to maintain the axis,
- the flow geometry becomes closed or partially closed.

Open vortex structures are unstable and dissolve quickly. Only vortices with closed axis geometry can develop into stable material vortices.

3.2 Stability Criteria for Material Vortices

A material vortex is a stable, closed vortex structure with quantized rotational tension. Its stability arises from the balance between the internal rotational tension and the external counteracting tension of the continuum.

The stability criteria are:

- **Closed axis:** The vortex axis must be fully closed to preserve the rotational geometry.
- **Quantized rotational tension:** The tension along the axis must assume a stable mode.
- **Flow coherence:** The flow lines of the medium must be coherent and synchronized.
- **Medium embedding:** The vortex must be embedded in the tension field of the continuum without destabilizing it.

Only vortices that fulfill all criteria can exist as stable material vortices. The first fully stable vortex is the proton vortex.

3.3 Radiative vs. Material Vortices

Two fundamental vortex types exist in the MCM:

- **Radiative vortices:** open, unstable vortices that emit energy and dissolve quickly. They possess no closed axis and no quantized tension.
- **Material vortices:** stable, closed vortices with quantized rotational tension. They do not radiate and form the basis of material structures.

Radiative vortices are short-lived and contribute to the tension dynamics of the continuum without producing stable structures. Material vortices, by contrast, are long-lived and form the foundation of all later material emergences.

3.4 Transition from Proto-Vortices to Stable Material Vortices

Proto-vortices are the first rotating tension structures in the continuum, but they are still unstable and open. The transition to stable material vortices occurs when a proto-vortex forms a closed axis and its rotational tension becomes quantized.

The transition occurs in three steps:

1. **Local tension concentration:** A fluctuation generates a rotating flow geometry.
2. **Axis formation:** The rotation stabilizes and forms a closed axis.
3. **Quantization:** The rotational tension assumes a stable mode and the vortex becomes material.

The first fully stable material vortex is the proton vortex. It forms the basis of all later material structures and is the primary stable vortex configuration in the MCM.

4 Proton Vortex as Stable Fundamental Structure

The proton vortex represents the first fully stable material vortex structure in the Monistic Continuum Model (MCM). While proto-vortices and radiative vortices remain open, short-lived tension phenomena, the proton vortex achieves a closed axis geometry and a quantized rotational tension that stabilizes it permanently. It therefore forms the ontological foundation of all visible matter in the universe.

4.1 Ontological Role of the Proton Vortex

In the MCM, the proton vortex is not a composite object but a single, closed vortex configuration of the continuum. It possesses a triple-locked axis structure that keeps its rotational tension stable and produces persistent flow coherence. This stability makes the proton vortex the fundamental material unit of the universe.

The ontological role of the proton vortex includes:

- the **first stable material structure** in the continuum,
- the **foundation of all later forms of matter**,
- the **source of quantized tension modes**,
- the **basis for atomic and cosmic stability**.

Thus, the proton vortex is not merely a building block of matter but the primary stable manifestation of the continuum itself.

4.2 Axis Coupling and Rotational Tension

The stability of the proton vortex arises from a triple axis coupling. This coupling produces a closed rotational geometry in which the tension flows of the medium are synchronized and locked. The rotational tension assumes a quantized mode that cannot be further increased or decreased. The axis coupling consists of:

- a **closed main axis** that defines the vortex,
- two **secondary coupling axes** that stabilize the flow geometry,
- a **quantized rotational tension** that maintains the vortex permanently.

This triple coupling prevents any form of radiation or energy loss. The proton vortex therefore remains stable and unchanged over cosmic timescales.

4.3 Why Protons Are Stable

The stability of the proton vortex results from the complete closure of its flow geometry. While radiative vortices possess open axes and emit energy, the axis of the proton vortex is fully closed and locked. This creates a tension minimum that energetically stabilizes the vortex.

Protons are stable because:

- their axis is fully closed,
- their rotational tension is quantized and immutable,
- no open flow channels exist,
- no radiation is possible,
- the medium energetically supports the vortex.

These properties make the proton vortex the most long-lived and stable vortex structure in the entire continuum.

4.4 Protons as the Foundation of Cosmic Matter

Because the proton vortex represents the first stable material structure, it forms the foundation of all later cosmic matter. Every form of visible matter—from atomic structures to stars and galaxies—is based on the stability and quantized tension modes of the proton vortex.

The cosmic significance of the proton vortex includes:

- the **formation of atomic structures** through coupling with external vortex modes,
- the **stabilization of macroscopic matter** through coherent tension modes,
- the **generation of cosmic structures** through large-scale vortex distributions,
- the **distinction from residual vortices**, which act as dark matter.

Thus, the proton vortex is the central stable vortex structure of the MCM and the foundation of all visible matter in the universe.

5 Residual Vortices as Dark Matter

Residual vortices represent, in the Monistic Continuum Model (MCM), those short-lived, neutral tension remnants that arise from incomplete or failed vortex-formation processes. They possess no closed axis, no quantized rotational tension, and no material core structure. Despite their instability and lack of radiation, they act on cosmic scales as tension sinks, which physics interprets as dark matter. Residual vortices are therefore not an independent substance but an emergent by-product of the vortex dynamics of the continuum.

Property	Residual Vortex (Dark Matter)
Ontological	✓
Core	0
Axis	–
Rotation	Local tension anomaly
Radiation	–
Matter	–
Spectral Signature	–
Detectable	– (only indirectly)
Stability	Short-lived, statistically stable
Geometry	Open, diffuse field
Coupling Ability	✓ (with tension field)

5.1 Definition of the Residual Vortex

A residual vortex is an open, unstable vortex structure that arises from a failed coupling process between flow geometries of the continuum. While stable material vortices possess a closed axis and quantized rotational tension, residual vortices lack these structures entirely. They are therefore non-material, non-radiating, and ontologically pure tension anomalies within the medium.

Residual vortices are:

- **open**: no closed axis,
- **neutral**: no charge or tension quantization,
- **unstable**: short lifetime, no self-stabilization,
- **non-radiating**: no radiative modes,
- **tension-based**: purely geometric anomalies.

These properties make residual vortices invisible and only indirectly observable— through their effect on the flow geometry of the medium.

5.2 Why Residual Vortices Do Not Radiate

Radiation in the MCM arises exclusively from open vortex structures that emit energy through non-closed flow channels. Residual vortices do possess open geometries, but they lack stable rotational modes capable of producing coherent radiation. Their flow lines are incoherent, short-lived, and non-quantized, so no defined radiation frequency can arise.

Residual vortices do not radiate because:

- no quantized rotational tension exists,
- no coherent flow geometry is formed,
- no closed axis is present,
- no stable modes capable of radiation exist.

Thus, residual vortices are completely invisible to electromagnetic measurement methods and can only be detected through their geometric effects.

5.3 Tension Sinks Instead of Radiation

Although residual vortices do not radiate, they produce a measurable effect in the continuum: they form local tension sinks. These tension sinks act on cosmic scales like gravitational potentials, as they distort the flow geometry of the medium and influence the motion of stable material vortices. The effect of a residual vortex is therefore not energetic but geometric. It alters the structure of the tension field and produces a minimal yet large-scale distortion that manifests in physics as “gravitational attraction.”

Tension sinks lead to:

- distortions of the flow geometry,
- deflections of photon vortices (gravitational lensing),
- stabilization of galactic rotation curves,
- formation of large-scale cosmic structures.

Thus, the MCM explains all observed effects of dark matter without introducing new particles or fields.

5.4 Lifetime and Cosmic Abundance

Residual vortices are short-lived because they lack the closed axis geometry that characterizes stable material vortices. Nevertheless, they arise in enormous numbers, since every coupling process in the continuum—successful or not—produces tension remnants. The cosmic abundance of residual vortices is therefore extremely high, and their cumulative effect shapes the large-scale structure of the universe.

The lifetime of an individual residual vortex is short, but the **statistical overall effect** is permanent. The continuous formation of new residual vortices compensates for their rapid decay, so the universe maintains a stable background field of tension sinks.

The cosmic abundance results from:

- the high number of failed coupling processes,
- the constant vortex dynamics of the medium,
- the absence of radiation, which does not accelerate decay,
- the summation of many small tension sinks.

This background field is what physics measures as dark matter: an invisible, non-radiating component observable exclusively through its geometric effects.

6 Effects of Residual Vortices on Cosmic Structures

Residual vortices are open, unstable tension anomalies of the continuum that possess no material core structure and emit no radiation. Despite their instability, they produce local tension sinks that act on cosmic scales like gravitational potentials. The summation of these tension sinks forms an invisible background field that significantly shapes the dynamics of cosmic structures. In physics, this effect is interpreted as dark matter. In the MCM, however, it does not arise from new particles but from the geometric influence of residual vortices.

6.1 Tension Sinks as Gravitational Potentials

Residual vortices generate local tension sinks in the continuum. These sinks distort the flow geometry of the medium and influence the motion of stable material vortices. Although residual vortices possess no mass, they act like gravitational potentials because material vortices always follow the tension gradient of the medium.

The effect of a residual vortex corresponds to:

- a local reduction of tension energy,
- a distortion of flow lines,
- a deflection of stable vortex structures,
- an effective gravitational potential.

Thus, the MCM explains the gravitational effects of dark matter without introducing hypothetical particles.

6.2 Rotation of Galaxies

Galaxy rotation curves show that the outer regions of galaxies rotate faster than visible matter alone would allow. In the MCM, this effect arises from the summation of residual vortices in the galactic halo. The many small tension sinks generate a large-scale background potential that stabilizes the rotation speed.

The flat rotation curves result from:

- the high number of residual vortices in the halo,
- the summation of their tension sinks,
- the large-scale distortion of the flow geometry,
- the stabilizing influence on outer material vortices.

Thus, the MCM provides a natural explanation for galactic rotation curves without requiring additional mass.

6.3 Gravitational Lensing Through Tension Fields

Residual vortices influence not only material vortices but also photon vortices. Since photon vortices follow the tension gradient of the medium, their propagation paths are distorted by tension sinks. This distortion manifests as gravitational lensing.

In the MCM, gravitational lensing arises from:

- local tension sinks,
- large-scale tension fields,
- distortions of the flow geometry,

- deflection of photon vortices.

Thus, the MCM explains both strong and weak gravitational lensing without assuming invisible mass.

6.4 Large-Scale Structure Formation

In the early universe, enormous numbers of residual vortices formed because the vortex dynamics of the medium were extremely active. The distribution of these vortices created a tension network that significantly influenced the formation of large-scale structures.

Residual vortices led to:

- the formation of cosmic filaments,
- the emergence of voids,
- the stabilization of galaxy clusters,
- the structuring of the cosmic web.

This tension network corresponds to the large-scale structures observed in cosmological surveys.

6.5 Stabilization of Cosmic Systems

Residual vortices contribute to the stabilization of cosmic systems by smoothing local tension gradients and harmonizing the flow geometry of the medium. Without this background tension, many cosmic systems would be unstable or would dissolve more rapidly.

Stabilizing effects include:

- the stabilization of galactic halos,
- the damping of local tension disturbances,
- the harmonization of flow geometry,
- the long-term stabilization of cosmic structures.

Residual vortices thus act as an invisible, tension-based stabilization system of the universe.

6.6 No Radiation, but Persistent Influence

Residual vortices do not radiate because they lack closed axis geometry and quantized rotational tension. Nevertheless, their influence persists over cosmic timescales because new residual vortices are constantly forming and their cumulative effect remains stable.

Their persistent influence results from:

- the high rate of formation,
- the summation of many small tension sinks,
- the absence of radiation, which prevents accelerated decay,
- the global distortion of the tension field.

Thus, the residual-vortex background field forms the basis of all cosmic dark-matter effects in the MCM.

7 Physical Parallels

Over the past decades, modern physics has developed a wide range of models to explain the nature of dark matter. Although these models are based on different theoretical foundations, they share central properties that lie remarkably close to the interpretation offered by the Monistic Continuum Model (MCM). In particular, the assumption of an invisible, non-radiating component that acts solely through gravity aligns with the MCM interpretation of residual vortices. The following sections present the most important physical models and highlight their proximity to the MCM perspective.

7.1 Cold Dark Matter (CDM)

The CDM model describes dark matter as a cold, non-relativistic substance that does not interact electromagnetically. This component is invisible and acts exclusively through gravity. CDM forms the foundation of the standard model of cosmology (Λ CDM) and successfully explains:

- flat rotation curves of galaxies,
- gravitational lensing,
- large-scale structure formation,
- stability of cosmic halos.

From the MCM perspective, CDM functionally corresponds to the tension sinks of residual vortices. Both are invisible, non-radiating, and act solely through geometric distortions of the cosmic tension field. The difference lies in the ontology: CDM postulates new particles, whereas the MCM describes purely geometric vortex anomalies.

7.2 Self-Interacting Dark Matter

Self-Interacting Dark Matter (SIDM) extends CDM by introducing internal interactions within the dark-matter sector. These models aim to explain certain cosmological fine structures, such as:

- the core-cusp problem,
- the distribution of satellite galaxies,
- the internal dynamics of halos.

In the MCM, no particles exist, but residual vortices possess an internal dynamics that influence one another. The interaction of residual vortices arises from the superposition of their tension sinks and functionally corresponds to the internal interactions described by SIDM. Physics thus describes a behavior that naturally emerges from vortex mechanics in the MCM.

7.3 Dark Radiation

Dark radiation refers to additional radiation-like components that are not electromagnetic and therefore remain invisible. They influence:

- the acoustic peaks of the cosmic microwave background,
- structure formation in the early universe,
- the effective number of relativistic degrees of freedom.

In the MCM, no dark radiation exists in the physical sense, but tension modulations of the continuum can produce globally measurable effects without generating photons. These modulations functionally correspond to the effects that physics attributes to dark radiation.

7.4 Neutrino-like Relics

Neutrino-like dark-matter models postulate extremely light, weakly interacting relics that emit almost no radiation and are visible only through gravity. These models explain:

- the existence of a cosmic neutrino background,
- the weak interaction with visible matter,
- the purely gravitational signature.

Residual vortices in the MCM possess similar properties: they are neutral, non-radiating, and visible only through geometric effects. Physics remains within a particle-based framework, whereas the MCM provides a continuous tension-based interpretation.

7.5 Why Physics Comes Close but Does Not Suffice

The physical models come remarkably close to the MCM interpretation of dark matter because they share central properties:

- invisibility,
- absence of electromagnetic radiation,
- purely gravitational influence,
- large-scale structural impact,
- stability over cosmic timescales.

However, the physical models are ontologically insufficient because they:

- postulate new particles or fields,
- lack a geometric foundation,
- do not account for the vortex mechanics of the medium,
- cannot derive the emergence of dark matter from the ground state.

The MCM provides a closed, continuous ontology in which dark matter arises as residual vortices of the universal medium—without new particles, without additional fields, and fully compatible with all cosmological observations.

8 Comparison: Physics vs. MCM

Modern physics and the Monistic Continuum Model (MCM) describe many cosmic phenomena with remarkable agreement, particularly the properties of dark matter: invisibility, absence of electromagnetic radiation, purely gravitational influence, and large-scale structural impact. Nevertheless, the two approaches differ fundamentally in their ontological foundations. While physics postulates new particles or fields, the MCM describes dark matter as geometric tension anomalies of the universal medium. The following sections present the central differences and parallels.

8.1 Ontology: Particles vs. Continuum

Physics is based on a dualistic ontology of particles and fields. Dark matter is considered an independent substance that may consist of yet undiscovered particles. These particles are assumed to:

- possess no electromagnetic interaction,
- act only through gravity,
- remain stable over cosmic timescales.

In contrast, the MCM is strictly monistic. It contains no particles and no separate fields. Instead, only a single continuum exists, whose tension dynamics give rise to all observable phenomena. Dark matter is not a substance but an emergent by-product:

- residual vortices as open tension anomalies,
- tension sinks instead of mass,
- geometric distortions instead of particle interactions.

Thus, the ontological foundations differ fundamentally, even though the observed effects are nearly identical.

8.2 Radiation: Photons vs. Tension Modulation

In physics, radiation arises from photons, understood as quantized excitations of electromagnetic fields. Dark matter does not radiate because it has no electromagnetic interaction.

In the MCM, photons do not exist as particles. Radiation is a modulation of the continuum's tension, produced by open vortex structures. Residual vortices do not radiate because:

- no closed axis exists,
- no quantized rotational tension is present,
- no coherent modulation is generated.

Both models agree that dark matter does not radiate, but the underlying reasons are ontologically different.

8.3 Stability: Particle Physics vs. Vortex Mechanics

Physics explains the stability of dark matter through:

- extremely weak interactions,
- high mass or low mass (depending on the model),
- symmetry principles or new conservation laws.

In the MCM, stability arises from vortex mechanics. Residual vortices are short-lived, but their **statistical overall effect** is stable because new vortices constantly form. The stability of cosmic dark-matter effects results from:

- the high formation rate of residual vortices,
- the summation of many tension sinks,
- the absence of radiation,
- the global distortion of the tension field.

This produces a persistent background field that functionally corresponds to the stable dark matter of physics.

8.4 Cosmology: CDM vs. MCM Tension Field

The Cold Dark Matter (CDM) model is the cosmological standard approach in physics. It explains:

- flat rotation curves,
- gravitational lensing,
- large-scale structure formation,

- stability of galactic halos.

In the MCM, the same effects arise from the tension field of residual vortices:

- tension sinks act like gravitational potentials,
- photon vortices follow tension gradients (lensing effect),
- residual vortices form a cosmic tension network,
- halos stabilize through the summation of many vortex anomalies.

CDM and MCM are therefore functionally almost identical, yet fundamentally different in their ontological interpretation.

9 Conclusion

The Monistic Continuum Model (MCM) provides a continuous and ontologically closed explanation for dark matter that does not require the introduction of new particles or additional fields. Instead of postulating an invisible substance, the MCM describes dark matter as residual tensions of the universal medium: open, unstable vortex remnants that possess no closed axis geometry and therefore do not radiate. These residual vortices are observable only through their geometric effects, which manifest on cosmic scales as tension sinks and functionally correspond to the gravitational potentials of physical dark-matter models.

Modern physics comes remarkably close to this interpretation. Models such as Cold Dark Matter, self-interacting dark matter, dark radiation, and neutrino-like relics describe an invisible, non-electromagnetically interacting component that is detectable only through gravitational effects and shapes the large-scale structure of the universe. These properties align directly with the effects of residual vortices in the MCM. The difference does not lie in the observed phenomena but in the ontological foundation: while physics assumes new particles, the MCM explains the same effects through the tension dynamics of a single medium.

Thus, the MCM offers an elegant, consistent, and fully continuum-based alternative to particle-based dark-matter theories. It integrates all relevant cosmological observations—from rotation curves and gravitational lensing to large-scale structure formation—into a single geometric framework. In the MCM, dark matter does not appear as an independent substance but as an unavoidable by-product of the vortex mechanics of the universal medium. This perspective closes the ontological gap present in physical models and connects cosmic dynamics with a unified, monistic foundational structure.

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10 Recommended Citation

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